

Topic Clustering Prompt
<Task Overview> Your task is to assign the following article to an existing topic cluster or create a new topic cluster. # Article to cluster # [Date: {event_date}] {article_text} # Additional Context # {structured_data} # Existing Clusters # {cluster_list} <Instructions> (1) Articles about the SAME EVENT/TOPIC → assign to existing cluster (2) Compare keywords: if 70%+ overlap → assign to that cluster (3) Articles about CLEARLY DIFFERENT topics → create new cluster (4) Same people/entities but different events → different clusters (5) Different dates OK if same ongoing story (6) Use both article text AND additional context to identify the topic (7) Extract up to 10 keywords that capture the topic (8) If assigning to existing cluster, update keywords accordingly <Answer Format> (cluster_id, [keywords])

Figure 7: Prompt template for topic clustering task.

Temporal QA Prompt
<Task Overview> Your task is to answer the following question based on the provided articles. # Additional Context (by date) # [{date_1}] {structured_data_1} [{date_2}] {structured_data_2} # Articles # Article 1: [Date: {date_1}] {article_text_1} Article 2: [Date: {date_2}] {article_text_2} # Question # {question} # Choices # a. {choice_a} b. {choice_b} c. {choice_c} d. {choice_d} <Instructions> (1) Read all articles carefully (2) Use the provided information to answer the question (3) Use the additional context when helpful (4) Respond with ONLY the letter (a, b, c, or d) Answer:

Figure 8: Prompt template for temporal question answering task.

Summarization Prompt

<Task Overview>
Your task is to create a concise summary that covers ALL major events from the following articles.

Articles
Article 1: [Date: {event_date_1}] {article_text_1} Article 2: [Date: {event_date_2}] {article_text_2}

Additional Context
Event 1: {structured_data_1} Event 2: {structured_data_2}

<Instructions>
(1) Identify ALL distinct events/topics in the articles
(2) Briefly mention each event - don't focus on only one
(3) Keep the summary concise and balanced
(4) Use the additional context to identify key entities when helpful
Summary:

Figure 9: Prompt template for multi-event summarization task.

CheckEval Prompt

System: You are an expert evaluator. Output ONLY valid JSON.

User:

Reference Summary # {reference}

Generated Summary # {generated}

Evaluation (Answer Yes or No for each)

- **faithfulness:** Does the summary contain only facts that are present in or inferable from the reference? (No hallucinated information)
- **coverage:** Does the summary cover the key information from the reference?
- **non_redundancy:** Does the summary avoid unnecessary repetition?
- **relevance:** Does the summary focus on the main topic and include only relevant information?
- **coherence:** Does the summary have logical flow and good organization?

Output as JSON only:
{ "faithfulness": "Yes/No", "coverage": "Yes/No", "non_redundancy": "Yes/No", "relevance": "Yes/No", "coherence": "Yes/No" }

Figure 10: Prompt template for LLM-as-Judge evaluation (CheckEval) using GPT-4o-mini.

Multi-Stage Structural Cue Generation

[Stage 1: Entity Extraction]

Your task is to identify specific types of entities mentioned explicitly in the provided event summary and text.

<Objectives>

- (1) **People:** Named individuals only. DO NOT include groups, titles without names, or organizations.
- (2) **Location:** Geographical places explicitly mentioned.
- (3) **Involved Organizations:** Organizations, companies, government bodies, institutions directly involved.

<Key Principles>

- (1) Extract ONLY information explicitly stated. If not mentioned, return an empty list [].
- (2) Distinguish clearly between individual people and organizational entities.

<Few-shot Example> {example_entity}

<Task> Event Summary: {event_summary} Event Text: {event_text}

[Stage 2: Event Attributes]

Your task is to determine the core characteristics of the event described in the summary and text.

<Objectives>

- (1) **Event Type:** Category of the event (Political, Environmental, Economic, Social, Conflict)
- (2) **Cause:** Direct trigger or reason for the event explicitly stated
- (3) **Effect:** Immediate outcome or consequence directly resulting from the event
- (4) **Action:** Specific actions or activities explicitly described
- (5) **Sentiment:** Overall tone (Positive, Negative, Neutral)

<Key Principles>

- (1) Extract ONLY information explicitly stated. If not mentioned, use null or empty string.
- (2) Ensure 'Effect' describes an immediate outcome, distinct from longer-term results.

<Few-shot Example> {example_attribute}

<Task> Event Summary: {event_summary} Event Text: {event_text}

[Stage 3: Outcomes & Stakeholders]

Your task is to identify the specific results of the event and the groups with a vested interest.

<Objectives>

- (1) **Result:** Key outcomes with prefixes ("Policy:" / "Impact:" / "Status:" / "Agenda:")
- (2) **Stakeholders:** Groups or categories with direct interest in the event's outcome

<Key Principles>

- (1) Extract ONLY information explicitly stated. If not mentioned, return an empty list [].
- (2) Stakeholders should represent groups with an interest, not just any mentioned organization.

<Few-shot Example> {example_outcome}

<Task> Event Summary: {event_summary} Event Text: {event_text}

Figure 11: Multi-stage pipeline for structured cue generation using GPT-4o. Each stage extracts specific information from event summaries and texts independently.

Multi-stage Temporal QA Generation and Verification

[Stage 1: QA Generation]

Your task is to generate a simple, clear question based on the given event metadata, following strict quality constraints.

<Question quality constraints>

- *Answer Concealment*: Do NOT include the answer target or lexically identical phrases from the Event Context.
- *Temporal Validity*: The question must target time-varying attributes (AVOID static attributes like birthplace, parentage) and be explicitly tied to a temporal reference (e.g., currently, after).

<Answer quality constraints>

- *Contextual Groundedness*: The answer must be supported by the Event Context.
- *Specificity Alignment*: The specificity of the answer must align with the specificity of the reported information.

Additional Context # {structural_cue}

Article # [Date: {date}] {article_text}

[Stage 2: Choice Generation]

You are a QA multiple-choice generator. Given a question and its correct answer, your task is to generate a final set of 4 high-quality answer choices for a multiple-choice question.

<Choice Quality Constraints>

- *Contextual Plausibility*: The choices must be plausible and relevant to the question context.
- *Specificity Consistency*: The choices must match the specificity level of the correct answer.
- *Mutual Exclusivity*: The selected choices must be mutually exclusive, with no redundancy or semantic overlap.

[Stage 3: Automated Verification]

Evaluate question quality based on two criteria:

- *Answer Concealment*: Question should not include the answer target.
- *Temporal Validity*: Question must target time-varying attributes or causal relationships and be explicitly tied to a temporal reference.

Evaluate choice quality based on two criteria:

- *Contextual Plausibility*: Choices must be contextually relevant and match the specificity level of the answer.
- *Mutual Exclusivity*: Choices should be mutually exclusive, ensuring no redundancy or semantic overlap.

Evaluate answer quality based on two criteria:

- *Contextual Groundedness*: Answer should be grounded in the Event Context.
- *Specificity Alignment*: The specificity of the answer should be aligned to the specificity of the reported information.

Return score 0-2 for each component (Question, Choices, Answer), with one point awarded per satisfied criterion.

Figure 12: Prompt template for multi-stage temporal QA generation and verification. The pipeline consists of three stages: (1) question generation under temporal and answer-concealment constraints, (2) multiple-choice generation with plausibility and mutual-exclusivity constraints, and (3) verification, where question, choices, and answer are scored against explicit criteria.